

Name: Dhinesh Kumar

Batch: 12 - 2

Front-end

JS - Test

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PART 4

(1)

Variable is a predefined

Primary keyword holds value or data. It has a predefined keyword,

name, assignment Operator and the value.

Variables are classified as var, let, Const.

In var we can declare, redeclare, assign and reassign. It is older way.

In let we can declare and assign but we can't redeclare.

In Const we can only declare, and we cannot able to reassign, redeclare.

(2)

Data type :

Data type stores value in predefined data type. Based on types the process will change. Data type has two types

→ Primitive

→ Non Primitive.

Examples for.

String = "Dhinesh" (Text)

Number = 20 (Number)

Boolean = "true" (True or False)

Undefined = let name; (declared but not assigned)

Nan = let name = null; (Empty value)

(3)

Diff. b/w == and ===

==	===
Loose Equality	Strict Equality
Compares value after of Possible Conversion	Compares both value and data type.
Type Conversion may happen	No type conversion

Example :

`(==)`

`Console.log(5 == "5");`

Output = True

Here, `==` treats two values as equal after converting the string to a number.

`(===)`

`Console.log(5 === "5");`

Output = False.

Here, `===` it checks both the value and type match. The string remains the string.

④ Arithmetic Operators:-

It is used to mathematical Calculation

Example ;.

`+, -, *, /, %, **`

Assignment Operator :-

It is used to assign or update the values.

Examples :-

$=$, $+=$, $-=$, $*=$, $/=$

Comparison Operator :-

It is used to compare two values and return a Boolean result.

Examples :-

$>$, $<$, $>=$, $<=$, $==$, $!=$

Logical Operator :-

Combine or reverse condition.

$\&$ (AND), $\|$ (OR), $!$ (NOT).

⑤

Array :-

Array stores multiple value
in Ordered Collection from index
value (0).

Example :-

```
let fruits = ["Apple", "Mango", "orange"];  
console.log(fruits[1]);
```

Our put = mango.

Object :-

Object stores data has key-value pairs

Example

```
let Student = { name: "Dhinesh", age: 20,  
  course: "Full Stack" };
```

```
console.log(Student.name);
```

Our put = Dhinesh.

Array Objects :-

Stores multiple Objects in a array.

Example

```
let Students = [ { name: "Dhinesh", age: 20 },  
                  { name: "Ajith", age: 55 } ];
```

```
console.log (Students[0].age);
```

Output = 20.

⑥

Function :-

It is a reusable block of code that performs a specific task.

Example :-

```
function add(a, b) { return a+b; }
```

```
function add(a, b) { return a+b; }
```

```
(console.log (add(10, 20)));
```

Output = 30.

Parameter :- A variable declared in function definition, such as a and b.

Argument :

The value passed to the function
Such as (10 and 20).

Return :-

Sends a result back from the
function using the return keyword.

PART - B.

⑦ Even or Odd :-

```
let number = 20;  
if (number % 2 === 0) { console.log('even'); }  
else { console.log('odd'); }
```

Out put : ~~odd~~ even

⑧ For loop - 1 to 50. (even numbers)

```
for (let a = 1; a <= 50; a++) {  
  if (a % 2 === 0) { console.log(a); }  
}
```

Out put, 2, 4, 6, 8, 10, 12, 50

(9)

Factorial :-

```
let number = 5;  
let factorial = 1;
```

```
for (let a = 1; a <= number; a++)  
{  
    factorial = factorial * a;  
}
```

```
console.log(factorial);
```

Our Ans = 120

(10)

Fibonacci Series :-

```
let first = 0;  
let second = 1;
```

```
let third = 0;
```

```
for (let a = 0; a < 8; a++) {
```

```
    let third = first + second;
```

```
    first = second;
```

```
    second = third;
```

```
    console.log(a);
```

⑩

Array - Print Even -

```
let numbers = [10, 15, 20, 25, 30, 35, 40];
```

```
for (let a = 0; a < numbers.length; a++)
```

```
{ if (numbers[a] % 2 === 0)
```

```
{ console.log(numbers[a]); }}
```

```
Output = 10, 20, 30, 40.
```

⑫

Array Total :

```
let marks = [80, 75, 90, 85, 70];
```

```
let total = 0;
```

```
for (let a = 0; a < marks.length; a++)
```

```
{ total = total + marks[a]; }
```

```
console.log("Total = ", total);
```

```
Output : Total = 400
```

13

```
let Student = { name: "Dhines",  
  age: 20,  
  Course: "Full Stack",  
  mark: 85 };
```

```
Console.log("Student Name: ", Student.name);
```

```
Console.log("Course: ", Student.Course);
```

```
Console.log("Mark: ", Student.mark);
```

Our Pur :-

Student Name : Dhinesh

Course : Full Stack

Mark : 85

14

```
let Students = [
```

```
{ name: "Arun", mark: 80 },
```

```
{ name: "Kumar", mark: 45 },
```

```
{ name: "Priya", mark: 90 },
```

```
{ name: "Ram", mark: 35 } ];
```

```
for (let a = 0 ; a < students.length; a++)  
{  
  if (students[a].mark >= 50) {  
    console.log(students[a].name);  
  }  
}
```

Output ::

Arun
Priya.

PART - D

(15) Largest number ::

```
function largest(a, b) {  
  if (a > b) {  
    return a; }  
  else { return b; }  
}
```

```
console.log(largest(40, 25));
```

Output = 40.